Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

A projectile fired with initial velocity u at some angle θ has a range R. If the initial velocity be doubled at the same angle of projection, then the range will be

(A) $2\mathrm{R}$

(B) $\mathrm{R}/2$

(C) R

(D) $4\mathrm{R}$

Q.2

A particle moves in a plane along an elliptic path given by $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$. At $(0,b)$ the x-component of velocity is u. The y-component of acceleration at this point is

(A) $\frac{-bu^2}{a^2}$

(B) $\frac{-u^2}{b}$

(C) $\frac{-au^2}{b^2}$

(D) $\frac{-u^2}{a}$

Q.3

A particle of mass m is projected from the ground with an initial speed u_0 at an angle α with the horizontal. At the highest point of its trajectory, it makes a completely inelastic collision with another identical particle, which was thrown vertically upward from the ground with the same initial speed u_0. The angle that the composite system makes with the horizontal immediately after the collision is

(A) $\frac{\pi}{4}$

(B) $\frac{\pi}{4}+\alpha$

(C) $\frac{\pi}{4}-\alpha$

(D) $\frac{\pi}{2}$

Q.4

If the range of a gun which fires a shell with muzzle speed V is R, then the angle of elevation of the gun is

(A) $\cos^{-1}\left(\frac{\mathrm{V}^2}{\mathrm{Rg}}\right)$

(B) $\cos^{-1}\left(\frac{\mathrm{gR}}{\mathrm{V}^2}\right)$

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(C) $\frac{1}{2}\left(\frac{\mathrm{V}^2}{\mathrm{Rg}}\right)$

(D) $\frac{1}{2}\sin^{-1}\left(\frac{gR}{V^2}\right)$

Q.5

The position vector of a particle $\vec{R}$ as a function of time is given by $\vec{R} = 4\sin(2\pi t)\hat{i} + 4\cos(2\pi t)\hat{j}$ where R is in meters, t is in seconds and $\hat{i}$ and $\hat{j}$ denote unit vectors along x and y-directions, respectively. Which one of the following statements is wrong for the motion of particle

(A) Magnitude of acceleration vector is $\frac{v^2}{R}$, where v is the velocity of particle

(B) Magnitude of the velocity of particle is $10\pi \mathrm{m/s}$

(C) Path of the particle is a circle of radius $2\ \mathrm{m}$

(D) Acceleration vector is along $-\vec{R}$

Q.6

A cricketer can throw a ball to a maximum horizontal distance of $100\ \mathrm{m}$. With the same effort, he throws the ball vertically upwards. The maximum height attained by the ball is -

(A) $100\ \mathrm{m}$

(B) $80\ \mathrm{m}$

(C) $60\ \mathrm{m}$

(D) $50\ \mathrm{m}$

Q.7

A stone is thrown at an angle θ to the horizontal reaches a maximum height H. Then the time of flight of stone will be

(A) $\sqrt{\frac{2\mathrm{H}}{\mathrm{g}}}$

(B) $2\sqrt{\frac{2\mathrm{H}}{\mathrm{g}}}$

(C) $\frac{2\sqrt{2\mathrm{H}\sin\theta}}{\mathrm{g}}$

(D) $\frac{\sqrt{2\mathrm{H}\sin\theta}}{\mathrm{g}}$

Q.8

A body of mass m is thrown upwards at an angle θ with the horizontal with velocity v. While rising up the velocity of the mass after t seconds will be

(A) $\sqrt{(\mathrm{v}\cos\theta)^2 + (\mathrm{v}\sin\theta)^2}$

(B) $\sqrt{(\mathrm{v}\cos\theta - \mathrm{v}\sin\theta)^2 - \mathrm{gt}}$

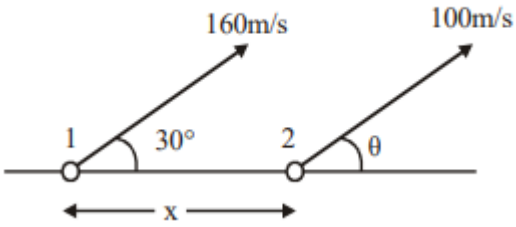
(C) $\sqrt{\mathrm{v}^2 + \mathrm{g}^2\mathrm{t}^2 - (2\mathrm{v}\sin\theta)\mathrm{gt}}$

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(D) $\sqrt{v^2 + g^2t^2 - (2v\cos\theta)gt}$

Q.9

Suppose two particles 1 and 2 are projected in vertical plane simultaneously.



Their angles of projection are 30° and θ, respectively, with the horizontal. Let they collide after a time t in air. Then -

(A) $\theta = \sin^{-1}(4/5)$ and they will have same speed just before the collision.

(B) $\theta = \sin^{-1}(4/5)$ and they will have different speed just before the collision.

(C) $x < 1280\sqrt{3} - 960$ m

(D) It is possible that the particles collide when both of them are at their highest point.

Q.10

Choose the correct options -

(A) If the initial velocity of a projectile be doubled, keeping the angle of projection same, the maximum height reached by it will be quadrupled

(B) The range of a projectile for a given initial velocity is maximum when the angle of projection is 45°. The range will be minimum, if the angle of projection is 90°.

(C) If the initial velocity of a projectile be doubled, keeping the angle of projection same, the maximum height reached by it will be double.

(D) The range of a projectile for a given initial velocity is maximum when the angle of projection is 45°. The range will be minimum, if the angle of projection is 30°.

Q.11

A projectile is fired from origin in xy-plane with speed 5 m/s at an angle ' θ ' with x-axis so as it hits the target at $(1, 1)$. Then the values of θ is/are $-$ $[g = 10 \text{ m/s}^2]$

(A) $\tan^{-1} 1$

(B) $\tan^{-1} 2$

(C) $\tan^{-1} 3$

(D) $\tan^{-1} 4$

Q.12

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Passage:
A balloon of mass $100\ \text{kg}$ with a basket of mass $25\ \text{kg}$ attached to it is ascending with a uniform upward velocity $8\ \text{m/s}$. A person of mass $50\ \text{kg}$ is also standing in the basket. He throws a packet of mass $2\ \text{kg}$ with an initial velocity $6\ \text{m/s}$ relative to him and perpendicular to the motion of the ascending balloon. The person observes that the packet hits the ground $4\ \text{s}$ after being thrown. Assume that the balloon continues to move with the same uniform velocity vertically upward and $g = 10\ \text{m/s}^2$,
Question: Height of the balloon when the packet hits the ground is nearly (initial relative velocity of ball wrt ground is given as 0)

(A) $92\ \text{m}$

(B) $72\ \text{m}$

(C) $80\ \text{m}$

(D) $112\ \text{m}$

Q.13

Passage:
A balloon of mass $100\ \text{kg}$ with a basket of mass $25\ \text{kg}$ attached to it is ascending with a uniform upnward velocity $8\ \text{m/s}$. A person of mass $50\ \text{kg}$ is also standing in the basket. He throws a packet of mass $2\ \text{kg}$ with an initial velocity $6\ \text{m/s}$ relative to him and perpendicular to the motion of the ascending balloon. The person observes that the packet hits the ground $4\ \text{s}$ after being thrown. Assume that the balloon continues to move with the same uniform velocity vertically upward and $g = 10\ \text{m/s}^2$,
Question: Let us set up a certensian coordinate system-take the initial position of the packet when it is thrown by the person as the origin, the vertically downward direction at the moment of throwing as the positive Y-direction and the initial direction of motion of the packet relative to the person as the positive X-direction. Coordinates of the packet when it hits the ground are (given that initial vertical component of velocity of packet relative to a person at rest is 0)

(A) $(16\hat{i} + 92\hat{j})\text{m}$

(B) $(24\hat{i} + 112\hat{j})\text{m}$

(C) $(24\hat{i} + 80\hat{j})\text{m}$

(D) $(12\hat{i} + 80\hat{j})\text{m}$

Q.14

Passage:
A balloon of mass $100\ \text{kg}$ with a basket of mass $25\ \text{kg}$ attached to it is ascending with a uniform upward velocity $8\ \text{m/s}$. A person of mass $50\ \text{kg}$ is also standing in the basket. He throws a packet of mass $2\ \text{kg}$ with an initial velocity $6\ \text{m/s}$ relative to him and perpendicular to the motion of the ascending balloon. The person observes that the packet hits the ground $4\ \text{s}$ after being thrown. Assume that the balloon continues to move with the same uniform velocity vertically upward and $g = 10\ \text{m/s}^2$,
Question: Velocity of the packet $1.6\ \text{s}$ after being thrown, relative to the person in the basket, has a magnitude nearly equal to (given that initial vertical component of velocity of packet relative to a person at rest is 0)

(A) $17\ \text{m/s}$

(B) $12\ \text{m/s}$

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(C) $21\ \mathrm{m/s}$

(D) $10\ \mathrm{m/s}$

Q.15

Passage:
A ball of mass $200\mathrm{gm}$ is thrown at an angle 45° above the horizontal at $\mathrm{t} = 0$. It hits a wall at a horizontal distance $15\ \mathrm{m}$ at a point $5\ \mathrm{m}$ above the point of projection. Take $\mathrm{g} = 10\ \mathrm{m/s^2}$, ignore air resistance.
Question: Magnitude of initial velocity of the ball is

(A) $8\ \mathrm{m/s}$

(B) $21\ \mathrm{m/s}$

(C) $12\ \mathrm{m/s}$

(D) $15\ \mathrm{m/s}$

Q.16

Passage:
A ball of mass $200\mathrm{gm}$ is thrown at an angle 45° above the horizontal at $\mathrm{t} = 0$. It hits a wall at a horizontal distance $15\ \mathrm{m}$ at a point $5\ \mathrm{m}$ above the point of projection. Take $\mathrm{g} = 10\ \mathrm{m/s^2}$, ignore air resistance.
Question: Magnitude of change of linear momentum of the ball between $\mathrm{t} = 0$ and $\mathrm{t} = \mathrm{t}'$, t' being the instant when the ball hits the wall, is nearly -

(A) $3.2\ \mathrm{kg\ m/s}$

(B) $2.8\ \mathrm{kg\ m/s}$

(C) $4.6\ \mathrm{kg\ m/s}$

(D) $1.4\ \mathrm{kg\ m/s}$

Q.17

Passage:
A ball of mass $200\mathrm{gm}$ is thrown at an angle 45° above the horizontal at $\mathrm{t} = 0$. It hits a wall at a horizontal distance $15\ \mathrm{m}$ at a point $5\ \mathrm{m}$ above the point of projection. Take $\mathrm{g} = 10\ \mathrm{m/s^2}$, ignore air resistance.
Question: It is given that the ball hits a wall at a horizontal distance $15\ \mathrm{m}$ at a point $5\ \mathrm{m}$ above the point of projection. If a second wall of height $(5 - \mathrm{dx})\mathrm{m}$ with $\mathrm{dx} \to 0$ has to be constructed such that the ball just misses to hit this wall, horizontal distance of this wall from the point of projection will be -

(A) $7.5\ \mathrm{m}$

(B) $12\ \mathrm{m}$

(C) $10.5\ \mathrm{m}$

(D) $6.5\ \mathrm{m}$

Q.18

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A ball is projected from the ground with velocity v such that its range is maximum then match the column

Column I	Column II
(A) Velocity at half of the maximum height	(p) $\frac{v}{2}$
(B) Velocity at the maximum height	(q) $\frac{v}{\sqrt{2}}$
(C) Change in its velocity when it returns to the ground	(r) $v\sqrt{2}$
(D) Average velocity when it reaches the maximum height	(s) $\frac{v}{2}\sqrt{\frac{5}{2}}$

(A) $A-(p)B-(r)C-(s)D-(q)$

(B) $A-(s)B-(r)C-(p)D-(q)$

(C) $A-(p)B-(q)C-(r)D-(s)$

(D) $A-(r)B-(s)C-(q)$

Q.19

A ball is projected upwards from the top of tower with a velocity $50\ \mathrm{m/s}$ making an angle 30° with the horizontal. The height of tower is $70\ \mathrm{m}$. After how many seconds from the instant of throwing will the ball reach the ground.

Q.20

Two particles P and Q move with constant velocities $v_1 = 2\ \mathrm{ms}^{-1}$ and $v_2 = 4\ \mathrm{ms}^{-1}$ along two mutually perpendicular straight lines towards the intersection point O. At moment $t = 0$, the particles were located at distances $\ell_1 = 12\ \mathrm{m}$ and $\ell_2 = 19\ \mathrm{m}$ from O, respectively. Find the time when they are nearest

Answers & Solutions

Q.1 Answer:

4R

Solution:

$\mathrm{R} = \frac{\mathrm{u}^2 \sin 2\theta}{\mathrm{g}} \therefore \mathrm{R} \propto \mathrm{u}^2$

If initial velocity be doubled then range will become four times.

Q.2 Answer:

$\frac{-bu^2}{a^2}$

Solution:

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$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

$u_x = u \quad \text{at } (0,b)$

$u_y = 0$

diff w.r.t to time

$\frac{2x}{a^2}\frac{dx}{dt}+\frac{2y}{b^2}\frac{dy}{dt}=0$

[At $(0,b)$]

Again diff. w.r.t to time

$\frac{2x}{a^2}\frac{d^2x}{dt^2}+\frac{2}{a^2}\left(\frac{dx}{dt}\right)^2+\frac{2y}{b^2}\frac{d^2y}{dt^2}+\frac{2}{b^2}\left(\frac{dy}{dt}\right)^2=0$

$\Rightarrow 0+\frac{2}{a^2}u^2+\frac{2}{b}a_y+0=0$

[At $(0,b)$]

y-component of acceleration at $(0,b)$ is $a_y=\frac{-b}{a^2}u^2$.

Q.3 Answer:

$\frac{\pi}{4}$

Solution:

$H=\frac{u_0^2\sin^2\alpha}{2g}$

Velocity at max height

$v_1=u_0\cos\alpha \quad \Rightarrow v_x=u_0\cos\alpha$

When thrown vertically upward

$v_2^2=u_0^2-2gH$

$\Rightarrow \quad v_2^2=u_0^2-2\times g\times\frac{u_0^2\sin^2\alpha}{2g}$

$\Rightarrow \quad v_2^2=u_0^2\cos^2\alpha$

$\Rightarrow \quad v_2=u_0\cos\alpha \Rightarrow v_y=u_0\cos\alpha$

$\Rightarrow \quad \tan\theta=\frac{v_y}{v_x}=1\Rightarrow\theta=45^\circ.$

$\Rightarrow \theta=\frac{\pi}{4}$

Q.4 Answer:

$\frac{1}{2}\sin^{-1}\left(\frac{gR}{V^2}\right)$

Solution:

$\mathrm{R}=\frac{\mathrm{v}^2\sin 2\theta}{\mathrm{g}}\Rightarrow\sin 2\theta=\frac{\mathrm{gR}}{\mathrm{v}^2}\Rightarrow\theta=\frac{1}{2}\sin^{-1}\left(\frac{\mathrm{gR}}{\mathrm{v}^2}\right)$

Q.5 Answer:

Magnitude of the velocity of particle is $10\pi\mathrm{m/s}$

Solution:

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According to the question,

The position vector of the particle is given by,

$\vec{R} = 4\sin(2\pi t)\hat{i} + 4\cos(2\pi t)\hat{j}$

Suppose,

$x = 4\sin(2\pi t)$ **.....(1)**

$y = 4\cos(2\pi t)$ **.....(2)**

On squaring and adding (1) and (2) we get

$x^2 + y^2 = (4)^2$

The above equation represents the circle of radius 4 **m.**

We know that velocity vector given by,

$$\vec{v} = \frac{d\vec{R}}{dt}$$

$$\therefore \vec{v} = \frac{d}{dt}(4\sin(2\pi t)\hat{i} + 4\cos(2\pi t)\hat{j})$$

Q.6 Answer:

50 m

Solution:

Maximum range $= \frac{u^2}{g_2} = 100$ m

Maximum height $= \frac{u^2}{2g} = \frac{100}{2} = 50$ m

Q.7 Answer:

$2\sqrt{\frac{2H}{g}}$

Solution:

$H = \frac{u^2 \sin^2\theta}{2g}$ **and** $T^2 = \frac{4u^2\sin^2\theta}{g^2} \Rightarrow T^2 = \frac{4u^2\sin^2\theta}{g^2}$

$\therefore \quad \frac{T^2}{H} = \frac{8}{g} \Rightarrow T = \sqrt{\frac{8H}{g}} = 2\sqrt{\frac{2H}{g}}$

Q.8 Answer:

$\sqrt{v^2 + g^2t^2 - (2v\sin\theta)gt}$

Solution:

Instantaneous velocity of rising mass after t **sec will be** $v_t = \sqrt{v_x^2 + v_y^2}$ **where** $v_X = v\cos\theta =$

Horizontal component of velocity

$v_y = v\sin\theta - gt =$ **Vertical component of velocity**

$v_t = \sqrt{(v\cos\theta)^2 + (v\sin\theta - gt)^2} = \sqrt{v^2 + g^2t^2 - 2v\sin\theta gt}$

Q.9 Answer:

$\theta = \sin^{-1}(4/5)$ and they will have different speed just before the collision.

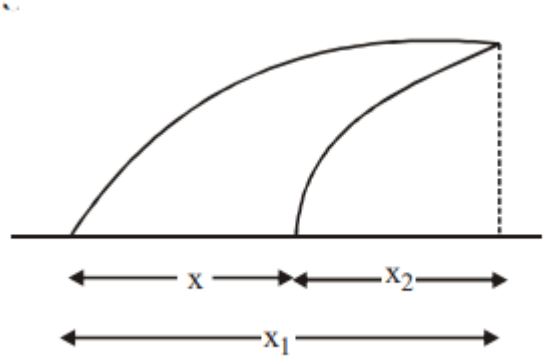
Solution:

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If they collide, their vertical component of velocities should be same i.e.,

$100 \sin\theta = 160 \sin 30^\circ \Rightarrow \sin\theta = 4/5$

Their vertical components will always be same.



Horizontal components :

$160 \cos 30^\circ = 80\sqrt{3}$ m/s **and**

$100 \cos\theta = 100 \times \frac{3}{5} = 60$ m/s

They are not same, hence their velocities will not be same at any time. So, (B) is correct.

$x = x_1 - x_2 = 160 \cos 30^\circ t - 100 \cos\theta t \Rightarrow x = (80\sqrt{3} - 60)t$

Time of flight : $T = \frac{2 \times 160 \times \sin 30}{g} = 16$ s

Now, $t < T_x \rightarrow$ **to collide in air** $\Rightarrow \frac{x}{80\sqrt{3}-60} < 16 \Rightarrow x < 1280\sqrt{3} - 960$

Since their times of flight are the same, they will simultaneously reach their maximum height.

So it is possible to collide at highest point for certain values of x.

Q.10 Answer:

If the initial velocity of a projectile be doubled, keeping the angle of projection same, the maximum height reached by it will be quadrupled

Solution:

(A) $H = \frac{u^2 \sin^2\theta}{2g} \therefore H \propto u^2.$

If initial velocity be doubled then maximum height reached by the projectile will quadrupled.

(B) Range $= \frac{u^2 \sin 2\theta}{g}$; **when** $\theta = 90^\circ, R = 0$

i.e. the body will fall at the point of projection after completing one dimensional motion under gravity.

Q.11 Answer:

$\tan^{-1} 2$

Solution:

$$y = x\tan\theta - \frac{1}{2} g \frac{x^2}{u^2 \cos^2\theta}; 1 = \tan\theta - \frac{5}{25}(1 + \tan^2\theta)$$

$$\Rightarrow \quad \tan^2\theta - 5\tan\theta + 6 = 0$$

$$\tan\theta = 2 \text{ or } \tan\theta = 3$$

$\Rightarrow \theta = \tan^{-1}(2)$ **or** $\tan^{-1}(3)$

Q.12 Answer:

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$112\ \mathrm{m}$

Solution:

velocity of ballon (upward) $= 8\ \mathrm{m/s}$

packet thrown prependicularly to ballon velocity so, the height of ballon at the time the packet was thrown

$\mathrm{H} = \mathrm{ut} + \frac{1}{2} g\ \mathrm{t}^2$

$\mathrm{u} = 0$ **(no velocity in vertical direction)**

$\mathrm{H} = \frac{1}{2} \times 10 \times 16 = 80\ \mathrm{m}$

the height travelled by ballon in $4\mathrm{sec} = 4 \times 8 = 32\ \mathrm{m}$

so total height of ballon at the time packet hit the ground w.r.t

ground $= 80 + 32 = 112\ \mathrm{m}$

Q.13 Answer:

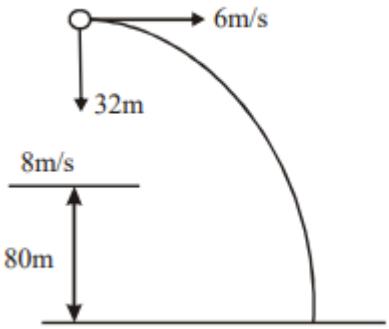
$(24\hat{\mathrm{i}} + 112\hat{\mathrm{j}})\mathrm{m}$

Solution:

Height of the balloon when the packet was thrown $= 80 + 8 \times 4 = 112\ \mathrm{m}$

$x = 6 \times 4 = 24\ \mathrm{m}$

Coordinate $= (24\hat{\mathrm{i}} + 112\hat{\mathrm{j}})\mathrm{m}$



Q.14 Answer:

$17\ \mathrm{m/s}$

Solution:

Relative to person $= \sqrt{6^2 + (1.6 \times \mathrm{g})^2} = \sqrt{36 + 256} = 17\ \mathrm{m/s}$

Q.15 Answer:

$15\ \mathrm{m/s}$

Solution:

$\mathrm{y} = \mathrm{x}\tan\theta - \frac{\mathrm{gx}^2}{2\mathrm{u}^2\cos^2\theta}$

$\Rightarrow 5 = 15 - \frac{10 \times 15 \times 15}{2 \times \mathrm{u}^2 \times \frac{1}{2}} \Rightarrow \mathrm{u} = 15\ \mathrm{m/s}$

Q.16 Answer:

$2.8\ \mathrm{kg\ m/s}$

Solution:

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$15 = 15\cos 45^\circ \cdot t$



$\frac{15}{\sqrt{2}}$

$t = \sqrt{2}\ s$

$v_y = \frac{15}{\sqrt{2}} - 10\sqrt{2} = -\frac{5}{\sqrt{2}}$



$\frac{5}{\sqrt{2}}$

$\Delta p = m\left(\frac{15}{\sqrt{2}} + \frac{5}{\sqrt{2}}\right) = \frac{200}{1000} \times \frac{20}{\sqrt{2}} = 2\sqrt{2} = 2.8\ kg\ m/s$

Q.17 Answer:

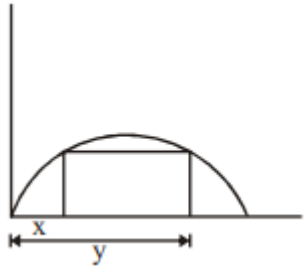
$7.5\ m$

Solution:

$x + y = R$

$R = \frac{15 \times 15}{10} = 22.5\ m$

$y = 15\ m, \quad x = 7.5\ m$



Q.18 Answer:

$A - (p) B - (q) C - (r) D - (s)$

Solution:

$(A) - \mathbf{p}, (\mathbf{B}) - \mathbf{q}, (\mathbf{C}) - \mathbf{r}, (\mathbf{D}) - \mathbf{s}$

(A) Range is maximum, when the angle of projection is $45^\circ\ H = \frac{v^2}{2g}\sin^2 45, H = \frac{v^2}{4g} ...(i)$

Velocity, at half of the maximum height is v'

$$v'^2 = v^2 \sin^2 45 - 2g\frac{H}{2}$$

$$v'^2 = \frac{v^2}{2} - \frac{v^2}{4} = \frac{v^2}{4} \Rightarrow v' = \frac{v}{2}$$

(B) Velocity at the maximum height $v' = v\cos 45^\circ \Rightarrow v' = \frac{v}{\sqrt{2}}$

[because vertical component of velocity is zero at the highest point]

(C) Projection velocity

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At projection point, $v_i = v\cos 45\hat{i} + v\sin 45\hat{j}$

At the point, when the body strike the ground $\vec{v}_f = v\cos 45\hat{i} - v\sin 45\hat{j}$

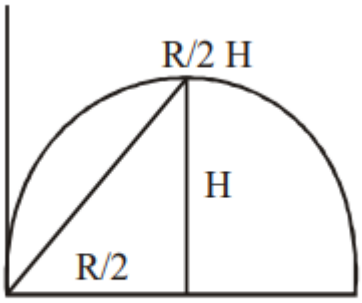
$$\Delta v = \vec{v}_f + \left(-\vec{v}_i\right) = 2\sin 45(-j)$$

$$|\Delta\vec{v}| = 2v\sin 45^\circ = v\sqrt{2}$$

(D) Average velocity $= \frac{\text{Total displacement}}{\text{Total time}}$ **Displacement** $= \sqrt{\left(\frac{R}{2}\right)^2 + H^2}$

$$\vec{v}_{av} = \frac{\sqrt{\frac{R^2}{4}+H^2}}{\frac{v\sin\theta}{g}} = \frac{\sqrt{R^2+4H^2}}{\frac{2v\sin\theta}{g}}$$

$$v_{av} = \sqrt{\frac{\left(\frac{v^2}{g}\right)^2+4\left(\frac{v^2}{4g}\right)^2}{\frac{\sqrt{2}v}{g}}} = \frac{\frac{v^2}{g}\sqrt{1+\frac{1}{4}}}{\frac{v\sqrt{2}}{8}} = \frac{v^2\sqrt{5}}{2v\sqrt{2}} = \frac{v}{2}\sqrt{\frac{5}{2}}$$



Q.19 Answer:

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Solution:

The vertical component of velocity of projection

If t **be the time taken to reach the ground,**

$h = ut + \frac{1}{2}gt^2 \Rightarrow 70 = -25t + \frac{1}{2}\times 10t^2 \Rightarrow 70 = -25t + 5t^2 \Rightarrow t^2 - 5t - 14 = 0$

$\Rightarrow t = -2$ s **and** 7 s**. Since,** $t = -2$ s **is not valid**

$\therefore \quad t = 7$ s

Q.20 Answer:

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Solution:

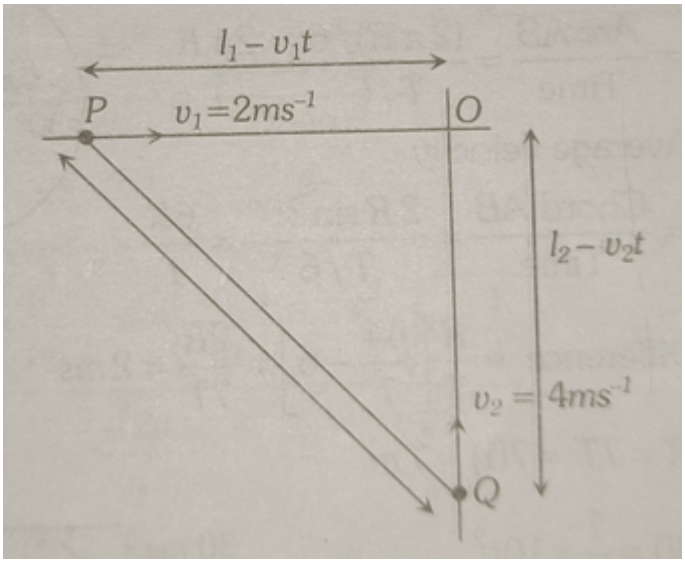
At any instant of time (t)

Distance $(S) = \sqrt{(l_1 - v_1 t)^2 + (l_2 - v_2 t)^2}$

For minimum S

$$\frac{ds}{dt} = 0 \Rightarrow 2s\frac{ds}{dt} = 2(l_1 - v_1 t)(-v_1) + 2(l_2 - v_2 t)(-v_2) = 0$$

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$$\Rightarrow \quad -I_1 v_1 + v_1^2 t - I_2 v_2 + v_2^2 t = 0$$

$$\Rightarrow \quad t = \frac{I_1 v_1 + I_2 v_2}{v_1^2 + v_2^2}$$

$$= \frac{12 \times 2 + 19 \times 4}{2^2 + 4^2} = 5\text{sec}$$

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